

Soumya Ranjan

AI/ML Engineer | ML Engineer | Data Engineer | Cloud ML Specialist

Bangalore, India

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Professional Summary

AI/ML Engineer with 5 years of experience in Python development and cloud-based system design, and 3+ years of hands-on experience building production-grade Machine Learning, Computer Vision, and Generative AI systems. Strong expertise in model development, cloud-based ML deployment (AWS & Azure), MLOps, and scalable backend systems. Proven ability to design end-to-end ML pipelines, optimize model performance, and deploy AI solutions at scale.

Technical Skills

Programming & Data

- Python (Advanced)
- SQL (MySQL, PostgreSQL)
- NoSQL (MongoDB, DynamoDB)
- Data Processing, Feature Engineering
- React js & Type script (front end)

Machine Learning & Modelling

- Supervised & Unsupervised Learning
- Deep Learning, Neural Networks
- Model Training, Evaluation & Optimization
- GPU-based Model Training

GPU & ML Frameworks

- TensorFlow
- Keras
- PyTorch
- MCP Framework

- OpenCV

Cloud-based ML Development & Deployment

- **AWS:** SageMaker, Bedrock, Lambda, S3, Glue
- **Azure:** Azure ML, Cloud-based ML deployment
- CI/CD for ML pipelines

MLOps & DevOps

- Model Deployment, Monitoring & Retraining
- Docker, Kubernetes
- GitHub, GitHub Actions, GoCD
- Terraform, Ansible

Databases & Storage

- MySQL, PostgreSQL
- MongoDB, DynamoDB
- Vector Databases: Qdrant, Pinecone

Professional Experience

AI/ML Engineer | Booster Netwave, Bhubaneswar

Mar 2023 – Present

- Designed, trained, and deployed ML and Deep Learning models for real-world production use cases by using PYTORCH, KERAS framework.
- Led multiple computer vision projects using YOLO-based architectures for real-time object detection.
- Built and deployed Generative AI applications using LLMs, RAG pipelines, and AI agents and building MCP framework for agents.
- Fine-tuned LLMs using QLoRA techniques, improving domain-specific performance by ~50%.
- Developed end-to-end ML pipelines using AWS SageMaker and MLOps best practices.
- Implemented monitoring and CI/CD pipelines for continuous model improvement.

- Designed data pipelines using Python, SQL, and NoSQL databases to support ML training and analytics.

Backend & System Design Engineer | Spark, Bangalore

Aug 2020 – Feb 2023

- Designed scalable backend architectures for high-traffic, cloud-native applications.
- Built microservices-based systems with REST, gRPC, and message queues.
- Implemented CI/CD pipelines and infrastructure as code for cloud deployments.
- Optimized system reliability and performance through architectural improvements.

Data Analyst Intern | Awign, Bangalore

Jan 2019 – Jun 2019

- Analyzed large datasets using Python and SQL to extract business insights.
- Built dashboards and reports for data-driven decision making.
- Performed data cleaning, preprocessing, and statistical analysis.

Key Projects

Intelligent AI Agent Framework

- Built autonomous AI agents using LLMs for customer service and internal automation.
- Implemented RAG pipelines and vector search for contextual responses.
- Tech: Python, GPT, QLoRA, LangChain & Lang graph, Vector Databases, MCP

Multi-Industry Computer Vision Platform

- Developed real-time computer vision systems for retail, healthcare, and manufacturing.
- Achieved 95%+ detection accuracy using YOLO-based models.
- Tech: Python, OpenCV, TensorFlow, AWS, Docker

Scalable ML Pipeline Architecture

- Designed MLOps infrastructure with automated training, deployment, and monitoring.

- Reduced model deployment time by ~60%.
 - Tech: AWS SageMaker, MLflow, Kubernetes, GitHub Actions
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Education

Master of Technology (MBA) – IT Management

VIT University, Vellore | 2019

Certifications & Achievements

- AWS Certified Solutions Architect (if applicable)
- Published research in Computer Vision applications
- Led cross-functional ML teams and reduced system latency by ~50%